PROJECT SUMMARY · OCTOBER 29, 2026

Meteo Alarm Kindle

How an old Kindle Paperwhite 1 became a weather clock with official weather warnings. The full story: every version, the city list, the favorites, and how the screen was designed.



Version 2.06 on the real Kindle, landscape mode, rain view, October 29, 2026.

3	25+	315	42	3
stages of work	builds and releases	towns in the city list	counties, plus Bucharest	favorite cities

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1. What the app is

Meteo Alarm Kindle turns a Kindle Paperwhite 1 (2012) into a weather clock. It runs on the Kindle itself. There is no server and no extra hardware. It needs only Wi-Fi.

- **Clock and date**, updated every minute without a flash.
- **Current weather:** big temperature, icon, description, "feels like", high/low and wind.
- **Forecast:** tomorrow, the day after, and three more days.
- **Rain or snow chart** for the next 12 hours, with a "now" marker.
- **Official warnings** (yellow, orange, red codes) for your county, from MeteoAlarm.
- **Sun, UV index and pollen** in landscape mode.
- **Front light:** a Kobo-style swipe on the left edge, with a 24-step light bar.
- **Portrait and landscape**, Romanian and English, 12h/24h, °C/°F.

Technically, it is a shell script started from KUAL (the Kindle Unified Application Launcher). FBInk draws on the e-ink screen. Weather, UV and pollen come from Open-Meteo. Warnings come from MeteoAlarm, issued in Romania by ANM.

2. How we worked

Viorel had the ideas, made the design decisions and ran every test on the real Kindle. Claude, Anthropic's AI assistant, wrote the code from those requests. Every feature went through the same loop:

1 Describe	2 Build	3 Preview	4 Test	5 Fix
Viorel asks for a feature	Claude writes the code	The screen is drawn on a PC	It runs on the real Kindle	Photos and logs show bugs

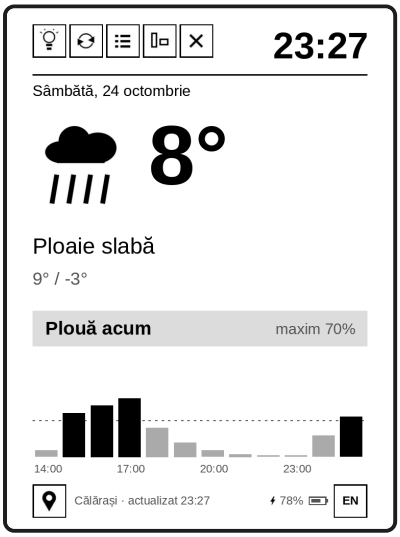
The loop repeated many times. Each round produced a new version.

Working rules that grew along the way

- **Questions get answers, not changes.** When Viorel asks "what can we do?", Claude only answers. Work starts only on a clear "let's start".
- **Ask before guessing.** If a request can be read two ways ("big letters": capitals or a larger size?), Claude asks first.
- **Clear version names.** Archives are named `MeteoAlarmKindle-vX.YY.zip`. 1.xx is the old screen, 2.xx the new one. A third digit (2.051) marks small design fixes.
- **PC preview first.** A PC simulator (a fake FBInk in Python, BusyBox, stubs for the Kindle tools) draws every screen before it reaches the Kindle.
- **Logs from the Kindle.** The log file has lines such as `weather:`, `warn:`, `pollen:` and `picker:`. They showed the cause of most bugs.
- **After every improvement:** update GitHub and add a comment on Reddit.

3. Stage 1: Kindle Weather Clock (v1 to 3.0)

October 24–25, 2026



The idea came from **rekindle**, an open-source clock and weather script for jailbroken Kindles, by Chaozhuang22 (MIT license). Seven builds turned it into a Romanian weather clock:

v1	Big clock and today's weather.
v2	5-day forecast.
v3	Touch buttons: light, flip, view, rotate, exit.
v4	Landscape mode.
v5	City picker, with the major Romanian cities.
v6	English/Romanian, 12h/24h clock, °C/°F.
v7 = 3.0	Final version, simpler light menu, larger buttons. Published as `kindle-weather-clock-ro.zip`.

Version 3.0: the first finished app.

4. Stage 2: accuracy and warnings (v31 to 1.35)

October 25–27, 2026

Viorel compared the Kindle with a Lenovo smart clock (Google weather). The goal: an icon and a temperature closer to the real sky.

v31	Icon from total cloud cover. At night it showed "cloudy" under a partly clear sky. Open-Meteo was 2° warmer than Google.
v32	Only low and mid clouds count; thin high clouds are ignored. A `MODEL` setting lets you pick the weather model. Icons now matched, including fog.
v33	The German DWD ICON model (`icon_seamless`) matched the temperature (14.2° vs 14°). Low clouds over 35% now mean "Overcast". "Feels like" and wind on their own lines. Battery status, "no data since" warning, automatic light at sunset, hourly weather at night.
v34	Refresh button and a Kobo-style light swipe on the left edge.
v35 = 1.35	Yellow / orange / red warnings from MeteoAlarm for your county. An `awk` error on the Kindle's old BusyBox hid them at first; it was fixed on October 27.



The Kindle next to the smart clock used as the reference.

Lessons from this stage:

- The Kindle has curl 7.33 and OpenSSL 1.0.1g. Plain HTTP always works; HTTPS works only sometimes. MeteoAlarm loads over HTTP.
- The Kindle's BusyBox 1.17 is stricter than a PC's. Code is now tested with a BusyBox build on the PC too.
- From here on, releases are named `MeteoAlarmKindle-vX.YY.zip`. v35 became **1.35**, the last version with the old screen.

Confirmed on the Kindle with 1.35: automatic light, hourly weather at night, light swipe, refresh, the 35% cloud rule and the battery status.

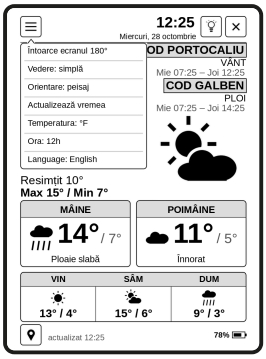
5. Stage 3: the new interface (2.01 to 2.061)

October 27–29, 2026

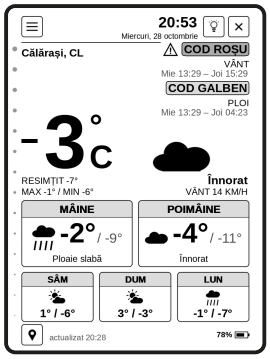
The 2.xx series is a new screen, designed in mockups first (see chapter 9) and then built in code. It draws at the Kindle's native resolution: 758×1024 in portrait, 1024×758 in landscape. Version 2.01 was tested on the Kindle; each later version fixed what the photos and logs showed.



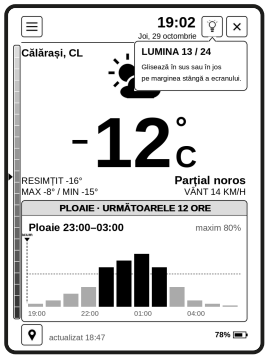
2.01



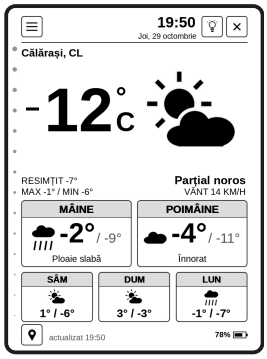
2.02



2.055



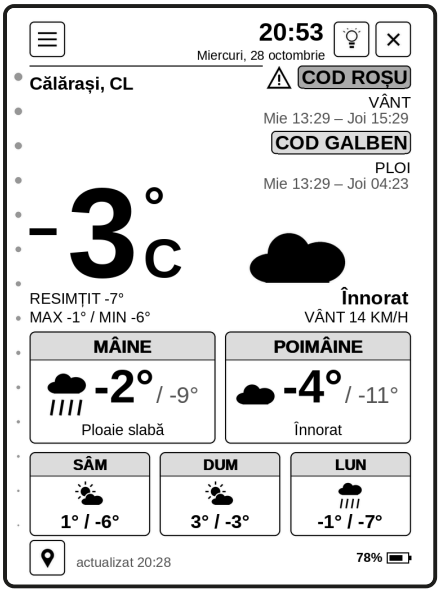
2.06



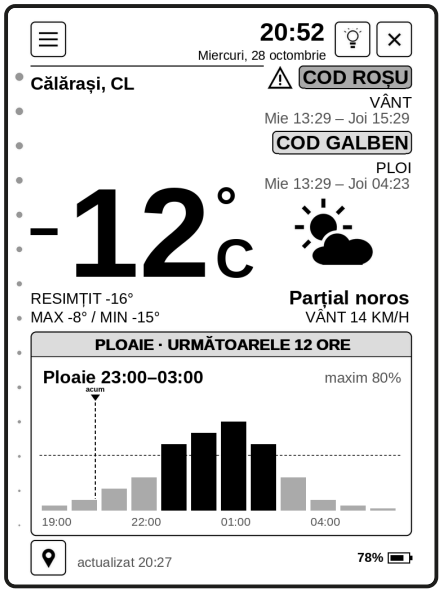
2.061

Version	What changed
2.01	First new interface. Rounded frames. Clock and date top right, then the light button and X. One ☰ menu with 7 rows (rotate 180°, view, orientation, refresh, °C/°F, 24h/12h, language). Light panel like the native Kindle (24 steps). City with county code ("Călărași, CL"). Up to two warnings at once. RAIN or SNOW box. Three day capsules when it is dry. Sun / UV / Pollen box in landscape.
2.02	Fixes after the first Kindle test. A broken top-left corner: text was measured by drawing it with FBInk. Text is now measured from glyph width tables. White text did not show, so gray bands now have black text. The menu became a speech bubble. Rain shown when it rains, even if the model says "cloudy". City list grew to its final size.
2.03	The city picker became a speech bubble too, with favorites on top and counties on two tabs.
2.04	A Back / Close / Next row at the bottom of the picker, as Viorel asked.
2.05	Gray dots mark the light swipe zone (60 px wide). Icon centered on the digits. Rounded warning bands. Light panel with MIN / MEDIU / MAX only.
2.051	Small design fixes: the warning box touches the line under the clock; three separate day boxes; icon 3% smaller in landscape.
2.052	New name: **Meteo Alarm Kindle** in KUAL. New folder `extensions/meteoalarmkindle`. On first start it copies the city and settings from the old rekindle folder.
2.053	The city picker opens at once (it used to redraw the whole screen first, 5–7 s). Favorites became the last three cities picked. Every tapped button turns medium gray. Larger light panel. Red code gets a "!" triangle.
2.054	With no warnings, the icon and the temperature share the free space, from Viorel's mockups. A thin, short minus sign. A "now" marker on the rain chart. Steadier pop-ups and a `CLOSE_OUTSIDE` setting.
2.055	Larger, bold box titles. Separate day boxes in landscape. RESIMȚIT, MAX / MIN and VÂNT in capitals. A new red-code triangle. The weather icon never covers the digits, the frame or the warning texts.
2.06	Pop-ups close at once: the screen under them is saved and put back. A Kobo-style light bar (24 steps) on the left edge, moved live by swiping. Battery messages no longer overlap.
2.061	Portrait, no warnings: temperature left (room for the minus), icon right. The last used ☰ menu choice moves to the top. No more dead menu after a long light swipe. Weather retried after 15 s.

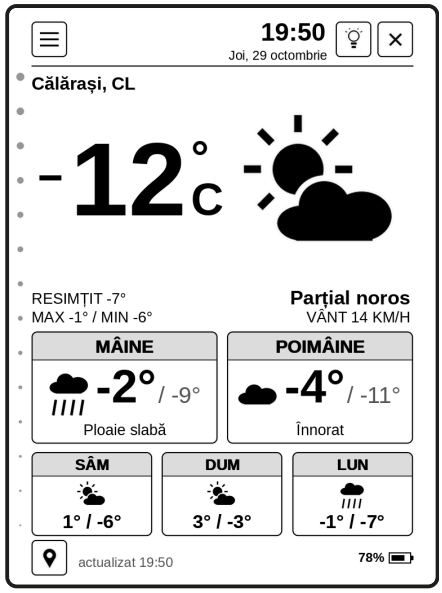
Version 2.061 today



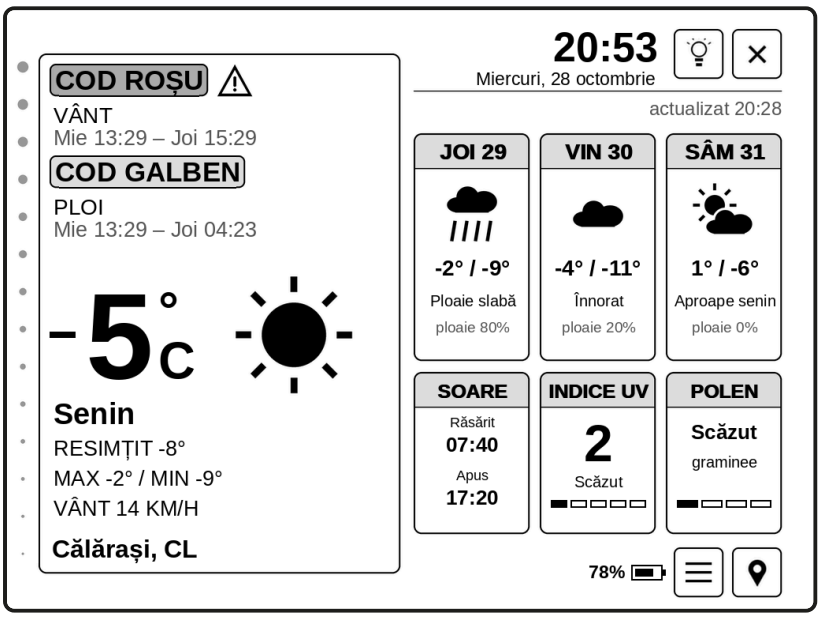
Portrait, 5 days, two warnings.



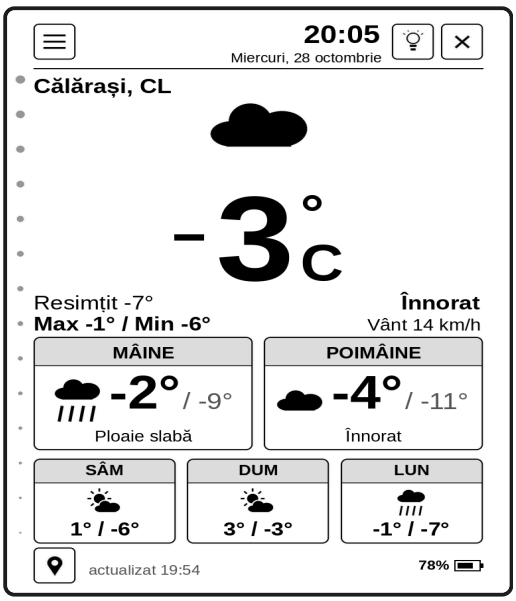
Portrait, rain chart with the "now" marker.



Portrait, no warnings, winter: temperature left, icon right (2.061).



Landscape: three day boxes and Sun / UV / Pollen.



A winter test screen with negative temperatures.

Screens are PC previews made by the app's own drawing code, except the photos.

6. The lock screen: no more ads



Viorel's Kindle (firmware 5.6.1.1) showed Amazon ads on the lock screen. The official way to remove them is no longer available. A small KUAL add-on, **ecranrepaus**, now does it. Version 1.3 was confirmed on the Kindle on October 28.

- It turns the ads off and empties the ad folder.
- It shows a Meteo Alarm Kindle screen instead of the ad.
- It runs by itself at every start, because the Kindle turned the ads back on after a restart.
- It fixes the clock: the Kindle was one hour ahead (stuck on summer time). It now uses Romanian time and changes to summer time by itself.
- On Viorel's Kindle its menu entry is hidden, so KUAL shows only Meteo Alarm Kindle. Do not delete its folder: the start-up job needs it.
- Published on October 29 as a separate project, with the menu shown: github.com/viorelrb-alt/meteo-alarm-kindle-screensaver-ro.

It took four rounds (1.0 to 1.3). Each round used the report file the add-on wrote on the Kindle, read over USB.

7. The city list

315	41	+1	2
towns	counties	Bucharest	starting favorites

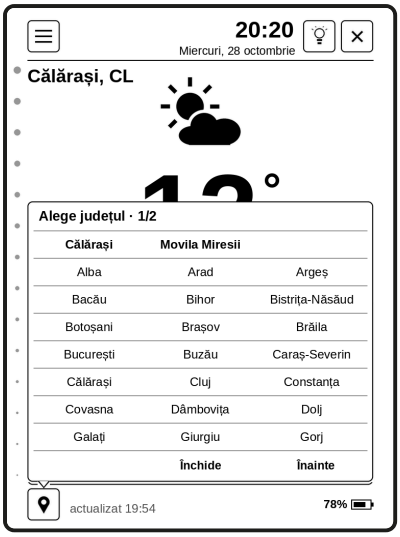
The list lives in `cities.txt`. I counted it in the 2.061 archive: **315 towns**, grouped by Romania's **41 counties plus Bucharest**. Two more lines are the starting favorites (Călărași and Movila Miresii). Earlier notes said 320; the file itself has 315 towns plus those 2 favorite lines, 317 lines in all.

- Stage 1 listed only the major cities. Version 2.02 merged in Viorel's own list of towns, checked against the old coordinates.
- Five old coordinates were kept, because the new list was off by 8 to 50 km: Lupeni, Cavnic, Corabia, Cehu Silvaniei and Videle.
- Each line is `county|town|latitude|longitude`. Anyone can add a town by copying a line.
- The county name is also used to filter the MeteoAlarm warnings, and the county code appears after the town ("Călărași, CL").

Towns per county

Alba	11	Covasna	5	Olt	8
Arad	10	Dâmbovița	7	Prahova	14
Argeș	7	Dolj	7	Satu Mare	6
Bacău	8	Galați	4	Sălaj	4
Bihor	10	Giurgiu	3	Sibiu	11
Bistrița-Năsăud	4	Gorj	8	Suceava	15
Botoșani	7	Harghita	8	Teleorman	5
Brașov	10	Hunedoara	13	Timiș	10
Brăila	5	Ialomița	7	Tulcea	5
București	1	Iași	5	Vaslui	5
Buzău	5	Ilfov	8	Vâlcea	10
Caraș-Severin	8	Maramureș	13	Vrancea	5
Călărași	5	Mehedinți	5	**Total**	315
Cluj	6	Mureș	11		
Constanța	11	Neamț	5		

8. How favorites work



Tap the pin button at the bottom of the screen. A speech bubble opens over the screen. Its **first row holds the favorites**: one tap and the city changes. Below them are the counties, on two tabs. Tap a county to see its towns, eight rows per page. The bottom row has **Back**, **Close** and **Next**. A tap outside the bubble closes it.

The rule

The favorites are **the last three cities you picked**, in this order:

1. the current city,
2. the one before it,
3. the one before that.

They are saved in `recent.sel` on the Kindle, so they survive a restart. The current city is also saved in `city.sel`.

An example

What you do	1st favorite	2nd	3rd
First start	Călărași	Movila Miresii	–
Pick Brașov	**Brașov**	Călărași	Movila Miresii
Pick Cluj-Napoca	**Cluj-Napoca**	Brașov	Călărași
Pick Călărași again	**Călărași**	Cluj-Napoca	Brașov

- **First start:** the list begins with the current city, then the lines marked `*` in `cities.txt`, with no repeats.
- **A city you pick** goes to the front. If it was already in the list, it moves up instead of appearing twice.
- **Only three are kept.** The oldest one drops off.
- Each favorite keeps its own coordinates, so a favorite can point to an exact spot, not only to the town center.

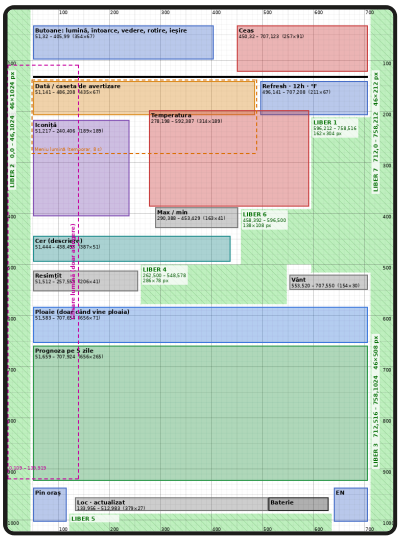
How the favorites came to be

In 1.35 the favorites were fixed lines in `cities.txt`. In 2.03 Călărași became a favorite with its own, more exact coordinates. The log from 2.052 showed a problem: the picker took 5–7 seconds to appear, so Viorel tapped the pin twice and the second tap closed it. Version 2.053 fixed this. The bubble now opens at once over the screen, taps during drawing are ignored, and the favorites became "the last three picked".

9. How we designed the screen

The design work moved from maps of the old screen, to mockups in Photon Studio, to code, and back to mockups whenever something needed a decision.

Step 1: a map of the old screen (October 26)

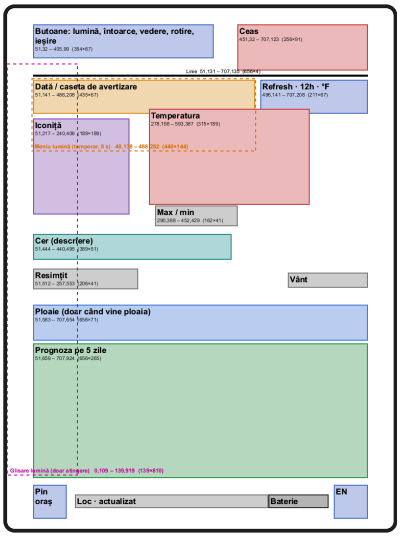


Claude drew maps of the 1.35 screen from the drawing code. Every element is a colored zone with its name and its pixel coordinates. Green marks the free space. Editable PSD and SVG versions followed, one layer per zone.

The maps showed two problems: a "-15°" temperature ran past the right edge in portrait, and the rain box touched "Wind" in landscape. Both were fixed in 2.01.

Step 2: mockups in Photon Studio (October 27)

Claude ran on Viorel's laptop through Remote Control and drew in Photon Studio. First came six zone mockups (portrait and landscape, three modes). Then a transparent layer called "Kindle screen v1.35 (example)", rendered on the PC with the app's own fonts, icons and gray levels, so Viorel could see the real screen over the zones.



Step 3: the new look (October 27)

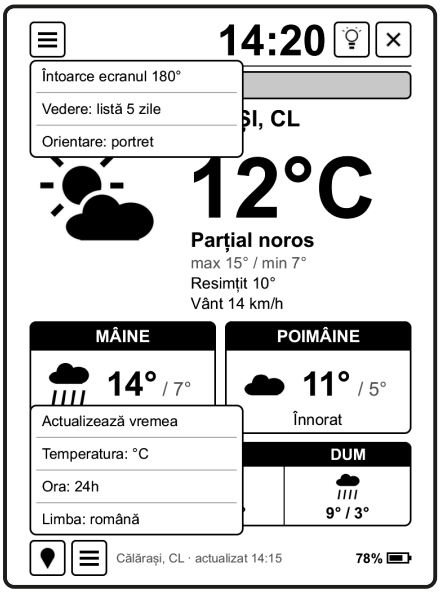


Viorel sent two example images of the style they wanted: rounded boxes with black outlines, black headers ("MÂINE", "POIMÂINE"), one large panel for the current weather and a row of three more days. Claude built the first v2 mockup from them, keeping the 1.35 weather icons.

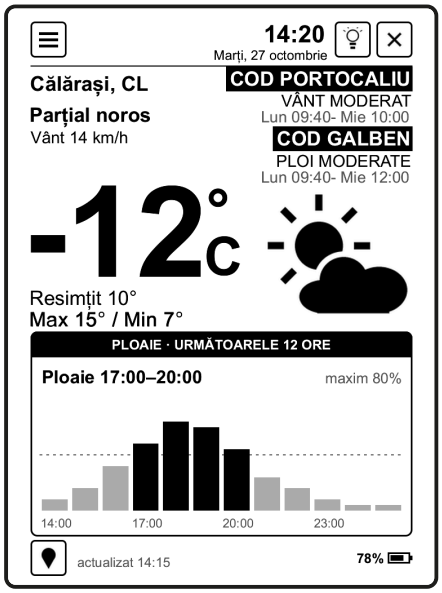
Viorel then edited the mockups personally in Photon. Claude checked what the Kindle could draw exactly and what only roughly. Rounded corners, for example, are made from small corner images, because FBInk draws only rectangles.

Step 4: decisions, one question at a time

Question	Viorel's decision
How many rows in the ☰ menu?	Seven. From 2.061 the order is orientation, view, rotate, refresh, °C/°F, 12h/24h, language, and the last used of the first four moves to the top.
What fills the space with no warnings?	First "FĂRĂ AVERTIZĂRI METEO"; from 2.054, the icon and the temperature.
Where does the light menu open?	First under the light button, like the native Paperwhite. From 2.06: a Kobo-style light bar on the left edge, set by swiping.
What does the simple mode show?	A rain chart when rain is coming. When it is dry: three separate day capsules (not warnings for three days).
What goes in landscape's free space?	A box with sunrise/sunset, UV index and pollen.
Is snow handled like rain?	Option 2: the box says SNOW or RAIN by the majority of hours, and snow bars are dark gray.



Mockup with the menus open.



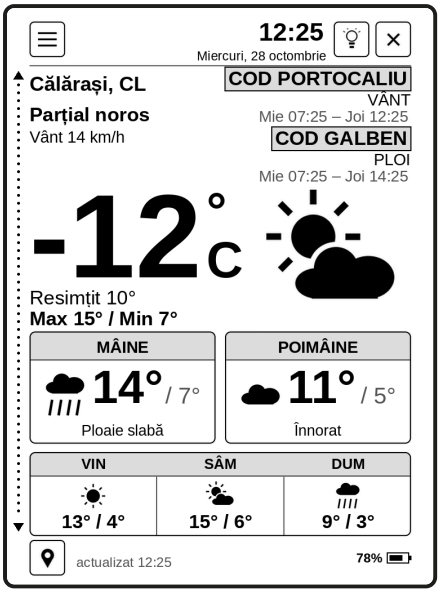
Simple mode with the rain chart.



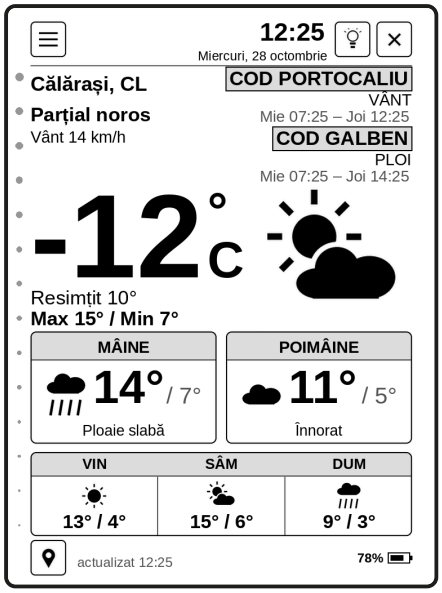
No warnings: icon above, temperature below.

Step 5: choosing between proposals

For smaller choices, Claude drew two or three versions side by side and Viorel picked one. The light swipe zone is a good example: dots with arrows (A), a gray strip (B), or dots that grow toward the top (B, chosen). The light panel was then shown at 120% next to the old size.



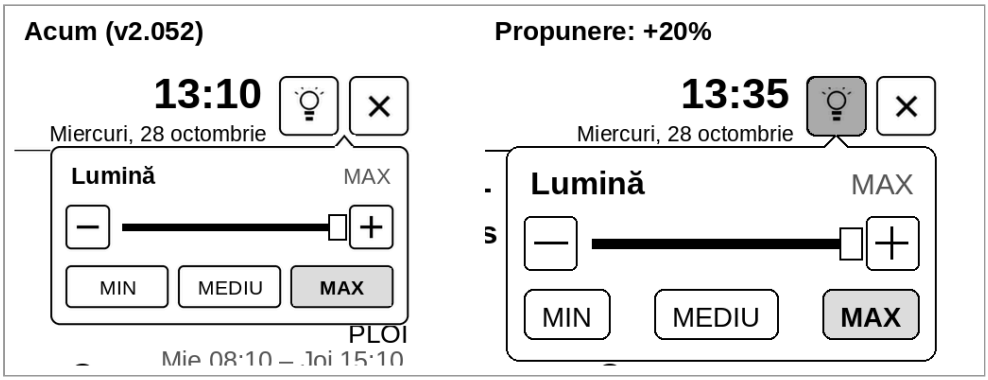
A: dots with arrows.



B: gray strip.



***Chosen:** dots that grow upward.*



The light panel, old size and 120%.

Step 6: Viorel draws the answer

When words were not enough, Viorel moved the elements in Photon directly. On October 28, Viorel edited four "no warnings" mockups to show exactly what they meant by "centered vertically and horizontally". The code in 2.054 follows those files. Viorel also asked for 18 winter screens with negative temperatures, and for larger, bold box titles, and approved the images before 2.055 was packed.

Feedback rounds in short

- Round 1 (2.01 on the Kindle): broken corner, invisible white text, menu style.
- Round 2 (2.02–2.04): city picker as a bubble, then a clear Back / Close / Next row.
- Round 3 (2.05–2.051): light zone marked, icon centered, warning boxes aligned with the line under the clock.
- Round 4 (2.052 log): slow picker and double taps, then the fixes in 2.053.
- Round 5 (2.054–2.055): calm-day layout, minus sign, larger titles, capitals for the info lines.
- Round 6 (2.06–2.061, October 29): faster pop-ups, a Kobo light bar, and a side-by-side layout that Viorel set in Photon.

10. Designing for a gray e-ink screen

The Paperwhite 1 shows 16 shades of gray at 212 pixels per inch. Many design choices come from that.

Limit	What we did
No color	Warning codes became shades. Yellow is light gray; orange and red are medium gray; red also gets a triangle with "!". Snow bars are dark gray, rain bars black.
White text did not show	The black headers of the mockups became light gray bands (0xDD) with black text.
Pressed buttons	A tapped button turns medium gray, not black, because the white text on black would vanish.
FBInk draws only rectangles	Rounded corners, bubbles and pill-shaped warning boxes are small PNG images, made on the PC with Python and Pillow.
Measuring text drew on the screen	Text widths come from tables made ahead of time (`fonts/*.wid`).
One font family	Everything uses Liberation Sans. "Bold" titles are drawn twice, 1 pixel apart.
E-ink ghosting	Closing a pop-up puts back the saved screen with one flash. A full refresh runs every 15 minutes. The clock updates without a flash.
Slow screen	Pop-ups are drawn straight over the screen, from images made ahead of time, so they open at once.
Readable from across the room	A huge temperature, a smaller "°" and "C", larger titles, and info lines in capitals.

11. What is still open

- Versions 2.055 and 2.06 ran on the Kindle on October 29. Version 2.061 was checked in PC previews and still needs a test on the Kindle, as do red and orange warnings, pollen, negative temperatures and battery use over 48 hours.
- An idea to discuss: redraw the screen only when the weather data changes, with a full refresh at least once an hour.
- GitHub: both projects were published on October 29, github.com/viorelrb-alt/meteo-alarm-kindle-ro and github.com/viorelrb-alt/meteo-alarm-kindle-screensaver-ro.

12. Credits

- **Ideas, design and testing:** Viorel.
- **Code:** Claude, Anthropic's AI assistant, from Viorel's requests.
- **Starting point:** rekindle by Chaozhuang22 (MIT license), github.com/Chaozhuang22/rekindle.
- **Jailbreak:** the Oak Reef guide; KUAL and tools from KindleModding and the MobileRead forums.
- **On the Kindle:** FBInk by NiLuJe; Liberation Sans (SIL Open Font License); BusyBox, curl, sqlite3 and upstart from the firmware.
- **Data:** Open-Meteo (CC BY 4.0) with the DWD ICON model; pollen from Copernicus CAMS; warnings from MeteoAlarm (EUMETNET), issued by ANM.
- **Design tools:** Photon Studio for the mockups; Python and Pillow for the PC previews.

Thank you to everyone who built and shared these tools and data for free.